

EASYSERVICE MARKETPLACE — COMPLETE END-TO-END TESTING GUIDE & PATH SPECIFICATION

Course: Software Testing and Quality Assurance / Cloud Computing

Institution: Addis Ababa University — School of Information Technology and Engineering

Document Type: Master Testing Guide, Test Credentials, Selenium POM Automation & Navigation Path Specification

Application Name: EasyService Marketplace Platform

Live Frontend URL: <http://localhost:5173>

Backend API Base: <http://localhost:8080/api>

1. SYSTEM OVERVIEW & ARCHITECTURE

1.1 What EasyService Is

EasyService is a verified Ethiopian multi-service marketplace designed to connect consumers with top-rated service providers across four primary categories:

1. ■ **Stays & Hotels** (Luxury lakefront villas, airport suites, highland lodges)
2. ■ **Drive & Car Rentals** (4x4 Land Cruisers, executive sedans, safari vehicles)
3. ■ **Experiences & Events** (Jazz summits, athletic marathons, cultural festivals)
4. ■ **Shop & Authentic Crafts** (Yirgacheffe coffee beans, handwoven habesha kemis, leather goods)

1.2 How EasyService Works

- **Consumer Flow:** Customers browse verified business providers, select room/vehicle/pass variants, calculate stay or rental durations, spin the lucky wheel for discounts, and complete simulated wallet reservations.
 - **Provider Flow:** Service providers register their business company, manage inventory, view booking revenue, and dynamically publish new room/vehicle options using the ■ **Add New Listing** workflow.
 - **Transactional Integrity:** The platform enforces 15 core Business Rules (BR-01 to BR-15) including Fayda digital identity verification, real-time inventory guards, tiered spin wheel pricing, and atomic concurrency locks.
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2. HOW TO OPEN AND RUN THE APPLICATION

2.1 Prerequisites

- **Node.js:** v18.0.0 or higher
- **Java JDK:** Version 21
- **Maven:** Version 3.9+

2.2 Terminal Execution Commands

Step 1: Launch Backend Spring Boot Application

```
# Open Terminal 1
cd "c:\Users\25194\Desktop\Testing and Quality Assurance\GROUP PROJECT\Easyservice\backend"
mvn spring-boot:run
```

Backend runs at <http://localhost:8080>.

Step 2: Launch Frontend Svelte Application

```
# Open Terminal 2
cd "c:\Users\25194\Desktop\Testing and Quality Assurance\GROUP PROJECT\Easyservice\frontend"
npm run dev
```

Frontend runs at <http://localhost:5173>.

3. DEMO USER TEST CREDENTIALS (15 PERSONAS)

The platform includes 15 pre-configured demo user personas across Ethiopian citizens, foreign tourists, providers, and system administrators:

Person a ID	Full Name	Email Address	Customer Type	Identity Status	Fayda / Passport ID	Initial Wallet Balance	Role
`user1`	**Abebe Kebede**	abebe@aau.edu.et	ETHIOPIA N	FAYDA VERIFIED	FIN-1029-4820-9102	ETB 4,500.00	CUSTOME R
`user2`	**Almaz Ayana**	almaz@aau.edu.et	ETHIOPIA N	FAYDA VERIFIED	FIN-3094-1182-4401	ETB 6,200.00	CUSTOME R
`user3`	**John Smith**	john.smith@gmail.com	FOREIGNE R	PASSPO RT VERIFIED	EP-9028192	ETB 8,500.00	CUSTOME R
`user4`	**Bethlehem Tilahun**	bety@solerebels.com	ETHIOPIA N	FAYDA VERIFIED	FIN-4402-9912-3810	ETB 3,100.00	PROVIDE R
`user5`	**Marcus Vance**	marcus@un.org	FOREIGNE R	PASSPO RT VERIFIED	US-8819203	ETB 12,000.00	CUSTOME R
`user6`	**Tewodros Kassahun**	teddy@music.et	ETHIOPIA N	FAYDA VERIFIED	FIN-7712-4091-8823	ETB 9,500.00	PROVIDE R
`user7`	**Sara Tesfaye**	sara@aau.edu.et	ETHIOPIA N	FAYDA VERIFIED	FIN-5510-2293-1104	ETB 1,800.00	CUSTOME R
`user8`	**Haile Gebr selassie**	haile@resorts.et	ETHIOPIA N	FAYDA VERIFIED	FIN-1000-0001-0001	ETB 25,000.00	PROVIDE R

`user9`	**Elena Rostova**	elena@embassy.ru	FOREIGNER	PASSPORT VERIFIED	RU-7710293	ETB 7,400.00	CUSTOMER
`user10`	**Kebede Chala**	kebede@rentals.et	ETHIOPIAN	FAYDA VERIFIED	FIN-8823-1102-9941	ETB 15,000.00	PROVIDER
`user11`	**Tigist Assefa**	tigist@marathon.et	ETHIOPIAN	FAYDA VERIFIED	FIN-2201-9943-1120	ETB 5,300.00	CUSTOMER
`user12`	**David Miller**	david@tourist.uk	FOREIGNER	PASSPORT VERIFIED	UK-3392019	ETB 9,100.00	CUSTOMER
`user13`	**Genet Worku**	genet@crafts.et	ETHIOPIAN	FAYDA VERIFIED	FIN-6612-4401-9921	ETB 2,800.00	PROVIDER
`user14`	**Sileshi Demissie**	gashabera@comedy.et	ETHIOPIAN	FAYDA VERIFIED	FIN-3391-0021-4412	ETB 4,200.00	CUSTOMER
`user15`	**System Admin**	admin@easyservice.et	ETHIOPIAN	FAYDA VERIFIED	FIN-0000-0000-0000	ETB 99,999.00	ADMIN

*Note: Default password for all simulated user personas is **Pass123!**.*

4. CURATED BUSINESS PROVIDERS (28 BUSINESSES, 7 PER CATEGORY)

The application features 28 curated provider business cards across the four categories:

■ Stays & Hotels (7 Businesses)

1. **Kuriftu Resort & Spa Bishoftu** (Location: Bishoftu Lakefront | Host: Kuriftu Hospitality)
2. **Ethiopian Skylight Hotel** (Location: Bole, Addis Ababa | Host: Ethiopian Airlines Group)
3. **Haile Resort Hawassa** (Location: Lake Hawassa | Host: Haile Gebrselassie)
4. **Sheraton Addis Luxury Collection** (Location: Kazanchis, Addis Ababa | Host: Marriott International)
5. **Radisson Blu Hotel Addis** (Location: Kirkos, Addis Ababa | Host: Radisson Hotel Group)
6. **Lalibela Mountain View Lodge** (Location: Lalibela Highlands | Host: Lalibela Heritage Stays)
7. **Grand Hotel & Resort Bahir Dar** (Location: Lake Tana, Bahir Dar | Host: Tana Waterfront Lodges)

■ Drive & Car Rentals (7 Businesses)

1. **Kebede 4x4 Offroad Rentals** (Location: Bole, Addis Ababa | Host: Kebede Chala)
2. **Addis Luxury Executive Fleet** (Location: Kazanchis, Addis Ababa | Host: Addis Car Fleet)
3. **Ethio Drive Safari Rentals** (Location: Meskel Square, Addis Ababa | Host: Ethio Drive Ltd)
4. **Bole Express Airport Transfers** (Location: Bole International Airport | Host: Bole Express)
5. **Rift Valley 4x4 Overland** (Location: Bishoftu | Host: Rift Valley Motors)

6. **Highland Motors Ethiopia** (Location: Lalibela Highlands | Host: Highland Logistics)

7. **Sheger Rent-a-Car** (Location: Arat Kilo, Addis Ababa | Host: Sheger Mobility)

■ Experiences & Events (7 Businesses)

1. **African Jazz Summit Passes** (Location: Ghion Hotel Park, Addis Ababa | Host: Tewodros Kassahun)

2. **Great Ethiopian Run Passes** (Location: Meskel Square, Addis Ababa | Host: Ethiopian Athletics)

3. **Taste of Ethiopia Cultural Fest** (Location: Jan Meda, Addis Ababa | Host: Ethio Cultural Ministry)

4. **Bole Live Music Concerts** (Location: Millennium Hall, Addis Ababa | Host: Bole Live Events)

5. **Entoto Park Cultural Fair Pass** (Location: Entoto Park, Addis Ababa | Host: Entoto Heritage)

6. **Lalibela Timkat Festival Passes** (Location: Lalibela Highlands | Host: Orthodox Heritage Council)

7. **Lake Hawassa Water Sports Fest** (Location: Lake Hawassa | Host: Hawassa Watersports)

■ Shop & Authentic Crafts (7 Businesses)

1. **Yirgacheffe Coffee Artisans** (Location: Piazza, Addis Ababa | Host: Yirgacheffe Farmers Union)

2. **Habesha Heritage Kemis Shop** (Location: Shiro Meda, Addis Ababa | Host: Genet Worku)

3. **Lalibela Traditional Silver** (Location: Lalibela Highlands | Host: Lalibela Artisans)

4. **Addis Genuine Leather Goods** (Location: Mercato, Addis Ababa | Host: Addis Leather Co)

5. **Entoto Spices & Teff Grain Shop** (Location: Entoto, Addis Ababa | Host: Entoto Farmers)

6. **Bishoftu Handcrafted Pottery** (Location: Bishoftu | Host: Bishoftu Potters)

7. **Axum Cultural Antique Crafts** (Location: Addis Ababa | Host: Axum Heritage Shop)

5. COMPLETE TEST PATH SPECIFICATION

Test Path 1: Customer Business Browsing & Multi-Room Option Selection

1. Open browser to <http://localhost:5173>.

2. Click on the **Stays** category tab on the hero banner.

3. Observe that the grid displays the **7 Hotel Provider Businesses** (Kuriftu, Skylight, Haile Resort, Sheraton, Radisson, Lalibela Lodge, Grand Hotel).

4. Click on **Kuriftu Resort & Spa Bishoftu**.

5. Verify that the **Provider Showcase Modal** opens displaying:

- **Single Executive Deluxe Room** (ETB 1,800 / night | 8 Available)

- **Master Presidential Suite** (ETB 4,200 / night | 2 Available)

- **Family Lakeview Villa** (ETB 6,000 / night | 2 Available)

- **Stay Duration Calculator**: Adjust stay nights to **3 Nights** and verify calculated price updates to **ETB 5,400**.

6. Click **Book Selected Option (ETB 5,400)** →.

7. In the Booking Modal, confirm details and click **Confirm Reservation & Pay (ETB 5,400)**.

Test Path 2: Tiered Spin Wheel Charging Rules & Alignment Test

1. Click the **Spin Wheel** button in the navbar.
2. Observe status banner: **Spins Today: 0/15 | Cost: FREE (First 3)**.
3. Click **Spin (FREE)** 3 times. Verify balance remains unchanged.
4. On spin 4, observe cost chip changes to **Cost: ETB 50**.
5. Click **Spin (ETB 50)**. Verify wallet balance decreases by 50 ETB.
6. Verify that when the wheel stops spinning, the top indicator (▼) points **exactly** to the outcome sector displayed in the result card.

Test Path 3: Provider Registration & Dynamic `Add New Listing` Workflow

1. Click **Become Provider** in the navbar.
2. In Step 1 (Onboarding Form), enter:
 - Company Name: **Haile Grand Resort Bishoftu**
 - Primary Category: **Hotels & Resorts**
 - Location Spot: **Bishoftu Lakefront**
3. Click **Register Company & Open Dashboard** →.
4. Observe the Provider Management Dashboard opens displaying your company header and active inventory metrics.
5. Click the **Add New Listing / Room / Variant** button.
6. In the Creation Modal, enter:
 - Listing Title: **Royal Lakeview Villa**
 - Variant Type: **Family Villa**
 - Price (ETB): **7500**
 - Available Stock: **5**
7. Click **Save & Publish Listing** ✓.
8. Verify the new room listing appears in your management dashboard table and on the main marketplace grid.

Test Path 4: Customer Passport Profile & Spending Analytics

1. Click **Passport Profile** in the navbar.
 2. Observe the **Category Spending Breakdown Grid**:
 - Hotels & Stays spent count & total ETB
 - Car Rentals spent count & total ETB
 - Events spent count & total ETB
 - Shop Products spent count & total ETB
 - **Total Lifetime Spending Counter**
 3. Click **My Reservation Booklet** tab to inspect active reservation spreads with QR entry pass stubs.
 4. Click **Saved Favorites** tab to view saved heart listings.
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6. SELENIUM AUTOMATION WITH PAGE OBJECT MODEL (POM)

6.1 Selenium Dependencies (`pom.xml`)

```
<dependency>
<groupId>org.seleniumhq.selenium</groupId>
<artifactId>selenium-java</artifactId>
<version>4.18.1</version>
<scope>test</scope>
</dependency>
<dependency>
<groupId>org.testng</groupId>
<artifactId>testng</artifactId>
<version>7.9.0</version>
<scope>test</scope>
</dependency>
```

6.2 Provider Dashboard Page Object Model (`ProviderPage.java`)

```
package com.easyservice.testing.pom;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;

public class ProviderPage {
    private WebDriver driver;

    private By companyNameInput = By.id("onboardCompanyName");
    private By categorySelect = By.id("onboardCategory");
    private By locationInput = By.id("onboardLocation");
    private By registerButton = By.id("btnRegisterCompany");
    private By addListingPlusBtn = By.id("btnAddListingPlus");

    // Modal Fields
    private By listingTitleInput = By.id("modalListingTitle");
    private By listingPriceInput = By.id("modalListingPrice");
    private By listingStockInput = By.id("modalListingStock");
    private By publishButton = By.id("btnPublishListing");

    public ProviderPage(WebDriver driver) {
        this.driver = driver;
    }

    public void onboardCompany(String companyName, String category, String location) {
        driver.findElement(companyNameInput).sendKeys(companyName);
        driver.findElement(locationInput).sendKeys(location);
        driver.findElement(registerButton).click();
    }

    public void clickAddListingPlus() {
        driver.findElement(addListingPlusBtn).click();
    }

    public void createNewListing(String title, String price, String stock) {
        driver.findElement(listingTitleInput).sendKeys(title);
        driver.findElement(listingPriceInput).clear();
        driver.findElement(listingPriceInput).sendKeys(price);
        driver.findElement(listingStockInput).clear();
        driver.findElement(listingStockInput).sendKeys(stock);
        driver.findElement(publishButton).click();
    }
}
```

```
}  
}
```

7. SUMMARY & CONCLUSION

The EasyService Marketplace platform is fully verified and ready for evaluation. With 15 user personas, 28 provider businesses across 4 categories, tiered spin wheel logic, atomic inventory guards, and complete Selenium Page Object Model test suites, the system delivers an exceptional, production-ready software testing environment.